# Double-Angle Formulas and Trigonometric Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.04

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Double-angle formulas

Given that $\cos\theta = -\frac{3}{5}$ and $\theta$ lies in Quadrant II, find:

**a** $\sin 2\theta = -\frac{24}{25}$ **b** $\cos 2\theta = -\frac{7}{25}$ **c** $\tan 2\theta = \frac{24}{7}$

(a) $= -\frac{24}{25}$; (b) $= -\frac{7}{25}$; (c) $= \frac{24}{7}$

**2** [Foundational] • Target: Double-angle formulas

Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.

$\cos^2 x = \frac{1+\cos 2x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$.

**3** [Foundational] • Target: Solving trigonometric equations

Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.

$\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$.

**4** [Foundational] • Target: Solving trigonometric equations

Solve $2\cos^2 x - \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

Factor: $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$.

**5** [Foundational] • Target: Solving trigonometric equations

Solve $\sin 2x = \sin x$ on the interval $[0, 2\pi)$.

$2\sin x\cos x - \sin x = 0$, so $\sin x(2\cos x - 1) = 0$. From $\sin x = 0$: $x = 0, \pi$; from $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}, \frac{5\pi}{3}$. Solution set: $\left\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$.

**6** [Foundational] • Target: Solving trigonometric equations

Solve $\tan^2 x = 3$ on the interval $[0, 2\pi)$.

$\tan x = \pm\sqrt{3}$, so $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

**7** [Foundational] • Target: Half-angle formulas

Use a half-angle formula to find the exact value of $\cos 15°$ (using $15° = \frac{30°}{2}$).

$\cos 15° = \sqrt{\frac{1+\cos 30°}{2}} = \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2+\sqrt{3}}{4}} = \frac{\sqrt{2+\sqrt{3}}}{2}$ (positive since $15°$ is in Quadrant I).

© ClipSAT — clipsat.org | All Rights Reserved

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: Half-angle formulas**

Given $\cos\theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin\frac{\theta}{2}$ exactly.

$\sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = \sqrt{\frac{18/25}{2}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$ (positive since $\frac{\theta}{2}$ is in Quadrant I).

**9** **[Intermediate]** • **Target: Equations requiring double-angle substitution**

Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

$1 - 2\sin^2 x + \sin x = 0$, so $2\sin^2 x - \sin x - 1 = 0$: $(2\sin x + 1)(\sin x - 1) = 0$. From $\sin x = -\frac{1}{2}$: $x = \frac{7\pi}{6}, \frac{11\pi}{6}$; from $\sin x = 1$: $x = \frac{\pi}{2}$. Solution set: $\left\{\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\right\}$.

**10** **[Intermediate]** • **Target: Equations requiring double-angle substitution**

Solve $\sin 2x - \cos x = 0$ on the interval $[0, 2\pi)$.

$2\sin x\cos x - \cos x = 0$, so $\cos x(2\sin x - 1) = 0$. From $\cos x = 0$: $x = \frac{\pi}{2}, \frac{3\pi}{2}$; from $\sin x = \frac{1}{2}$: $x = \frac{\pi}{6}, \frac{5\pi}{6}$. Solution set: $\left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\right\}$.

**11** **[Intermediate]** • **Target: Applications of double-angle formulas**

A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45°$.

$R(\theta) = \frac{v^2}{g}\sin 2\theta$ is maximized when $\sin 2\theta = 1$, i.e. $2\theta = 90°$, so $\theta = 45°$.

**12** **[Intermediate]** • **Target: Applications of double-angle formulas**

Given $\sin\theta = \frac{1}{3}$ with $\theta$ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

$\cos\theta = \sqrt{1 - 1/9} = \frac{2\sqrt{2}}{3}$. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{2}{9} = \frac{7}{9}$. $\sin 2\theta = 2\sin\theta\cos\theta = 2 \cdot \frac{1}{3} \cdot \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{9}$.

**13** **[Intermediate]** • **Target: More equation practice**

Solve $4\sin^2 x - 3 = 0$ on the interval $[0, 2\pi)$.

$\sin^2 x = \frac{3}{4}$, so $\sin x = \pm\frac{\sqrt{3}}{2}$: reference angle $\frac{\pi}{3}$, giving $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

**14** **[Intermediate]** • **Target: More equation practice**

Solve $\cos 2x = \cos x$ on the interval $[0, 2\pi)$, using $\cos 2x = 2\cos^2 x - 1$.

$2\cos^2 x - 1 = \cos x$: $2\cos^2 x - \cos x - 1 = 0$, so $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$.

© ClipSAT — clipsat.org | All Rights Reserved

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Half-angle formulas for tangent** • **Cross-Lesson: 8.03 Introduction to Limits**

State the half-angle formula for $\tan\frac{\theta}{2}$ in terms of $\sin\theta$ and $\cos\theta$ (the form avoiding a square root), and use it to find $\tan 15°$.

$\tan\frac{\theta}{2} = \frac{\sin\theta}{1+\cos\theta}$. With $\theta = 30°$: $\tan 15° = \frac{\sin 30°}{1+\cos 30°} = \frac{1/2}{1+\sqrt{3}/2} = \frac{1}{2+\sqrt{3}} = 2-\sqrt{3}$ (after rationalizing).

**16** **[Challenging]** • **Target: Equations with multiple angle types** • **Cross-Lesson: 9.01 Arithmetic Sequences and Series**

Solve $\sin 2x + \sin x = 0$ on $[0, 2\pi)$.

$2\sin x\cos x + \sin x = 0 \Rightarrow \sin x(2\cos x + 1) = 0$. From $\sin x = 0$: $x = 0, \pi$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$. Solution set: $\left\{0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}\right\}$.

**17** **[Challenging]** • **Target: Equations with multiple angle types** • **Cross-Lesson: 9.02 Geometric Sequences and Series**

Solve $\cos 2x = 1 - 3\sin x$ on $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.

$1 - 2\sin^2 x = 1 - 3\sin x \Rightarrow -2\sin^2 x + 3\sin x = 0 \Rightarrow \sin x(3 - 2\sin x) = 0$. From $\sin x = 0$: $x = 0, \pi$. From $\sin x = \frac{3}{2}$: no solution (out of range). Solution set: $\{0, \pi\}$.

**18** **[Challenging]** • **Target: Triple-angle preview** • **Cross-Lesson: 9.03 The Binomial Theorem**

Using $\sin(2x + x) = \sin 2x\cos x + \cos 2x\sin x$ together with the double-angle formulas, derive the triple-angle formula $\sin 3x = 3\sin x - 4\sin^3 x$.

$\sin 3x = \sin 2x\cos x + \cos 2x\sin x = (2\sin x\cos x)\cos x + (1 - 2\sin^2 x)\sin x = 2\sin x\cos^2 x + \sin x - 2\sin^3 x$. Substitute $\cos^2 x = 1 - \sin^2 x$:
$= 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x = 2\sin x - 2\sin^3 x + \sin x - 2\sin^3 x = 3\sin x - 4\sin^3 x$.

**19** **[Challenging]** • **Target: Graphical verification of a double-angle identity** • **Cross-Lesson: 1.01 Functions and Function Notation**

The functions $y = \cos 2x$ and $y = 1 - 2\sin^2 x$ are graphed together below. Explain what it means that the two curves appear identical, and state the identity this confirms.

The two curves overlapping perfectly (indistinguishable) is a visual confirmation of the identity $\cos 2x = 1 - 2\sin^2 x$ — since both functions produce the exact same output for every input $x$, their graphs coincide everywhere.

**20** **[Challenging]** • **Target: Solving equations restricted to a smaller interval** • **Cross-Lesson: 1.02 Domain and Range**

Solve $2\cos 2x - 1 = 0$ for $x \in [0, \pi)$ only (note the restricted domain).

$\cos 2x = \frac{1}{2}$: over $[0, 2\pi)$ for $2x$, solutions are $2x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$, giving $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$. Restricting to $x \in [0, \pi)$: only $x = \frac{\pi}{6}, \frac{5\pi}{6}$ qualify.

© ClipSAT — clipsat.org | All Rights Reserved

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Double-Angle Formulas

State the double-angle formula for $\sin(2\theta)$.

**a.** $2\sin\theta\cos\theta$ ✓

**b.** $\cos^2\theta - \sin^2\theta$

**c.** $2\sin\theta$

**d.** $\sin^2\theta$

$\sin(2\theta) = \sin(\theta+\theta) = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$.

**22** [Foundational] • Target: Double-Angle Formulas

Which of the following is NOT an equivalent form of $\cos(2\theta)$?

**a.** $1 - 2\cos^2\theta$ ✓

**b.** $\cos^2\theta - \sin^2\theta$

**c.** $2\cos^2\theta - 1$

**d.** $1 - 2\sin^2\theta$

The correct forms are $\cos^2\theta - \sin^2\theta$, $2\cos^2\theta - 1$, and $1 - 2\sin^2\theta$. $1 - 2\cos^2\theta = -\cos(2\theta)$.

**23** [Foundational] • Target: Double-Angle Formulas

If $\cos\theta = 4/5$ with $\theta$ in Quadrant IV, find $\sin(2\theta)$.

**a.** $-24/25$ ✓

**b.** $24/25$

**c.** $-7/25$

**d.** $7/25$

In QIV, $\sin\theta = -3/5$. $\sin(2\theta) = 2\sin\theta\cos\theta = 2(-3/5)(4/5) = -24/25$.

**24** [Foundational] • Target: Half-Angle and Power-Reducing Formulas

State the power-reducing formula for $\sin^2 x$.

**a.** $\frac{1-\cos(2x)}{2}$ ✓

**b.** $\frac{1+\cos(2x)}{2}$

**c.** $\frac{1-\sin(2x)}{2}$

**d.** $1 - \cos^2 x$

From $\cos(2x) = 1 - 2\sin^2 x$, solving for $\sin^2 x$ gives $\sin^2 x = \frac{1-\cos(2x)}{2}$.

© ClipSAT — clipsat.org | All Rights Reserved

**25** [Foundational] • Target: Half-Angle and Power-Reducing Formulas

Find the exact value of $\cos(22.5^\circ)$ using the half-angle formula $\cos(\theta/2) = \sqrt{\frac{1+\cos\theta}{2}}$.

**a.** $\frac{\sqrt{2+\sqrt{2}}}{2}$ ✓

**b.** $\frac{\sqrt{2-\sqrt{2}}}{2}$

**c.** $\frac{\sqrt{2}+1}{2}$

**d.** $\frac{\sqrt{3}+1}{2}$

$\cos(45^\circ/2) = \sqrt{\frac{1+\sqrt{2}/2}{2}} = \sqrt{\frac{2+\sqrt{2}}{4}} = \frac{\sqrt{2+\sqrt{2}}}{2}$.

**26** [Foundational] • Target: Half-Angle and Power-Reducing Formulas

Rewrite $\cos^4 x$ in terms of first powers of cosines of multiple angles.

**a.** $\frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)$ ✓

**b.** $\frac{1}{4}(1+\cos(2x))^2$

**c.** $\frac{1}{2} + \frac{1}{2}\cos(4x)$

**d.** $\frac{3}{8} - \frac{1}{2}\cos(2x)$

$\left(\frac{1+\cos 2x}{2}\right)^2 = \frac{1+2\cos 2x+\cos^2 2x}{4} = \frac{1+2\cos 2x+(1+\cos 4x)/2}{4} = \frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)$.

**27** [Foundational] • Target: Solving Linear Trigonometric Equations

Solve $\sqrt{3}\tan x + 1 = 0$ on $[0, 2\pi)$.

**a.** $\frac{5\pi}{6}$ **and** $\frac{11\pi}{6}$ ✓

**b.** $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

**c.** $\frac{\pi}{6}$ and $\frac{7\pi}{6}$

**d.** $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

$\tan x = -1/\sqrt{3}$. Tangent is negative in Quadrants II and IV at $x = 5\pi/6$ and $11\pi/6$.

© ClipSAT — clipsat.org | All Rights Reserved

**28** [Foundational] • Target: Solving Linear Trigonometric Equations

Find all solutions to $\cos(2x) = 0$ on $[0, \pi)$.

**a.** $\frac{\pi}{4}$ **and** $\frac{3\pi}{4}$ ✓

**b.** $\frac{\pi}{2}$

**c.** $\frac{\pi}{4}$ only

**d.** $\frac{\pi}{2}$ and $\frac{3\pi}{2}$

$2x = \frac{\pi}{2} + k\pi \Rightarrow x = \frac{\pi}{4} + \frac{k\pi}{2}$. On $[0, \pi)$, solutions are $\pi/4$ and $3\pi/4$.

---

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

Solve $2\sin^2 x - \sin x - 1 = 0$ on $[0, 2\pi)$.

**a.** $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

**b.** $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

**c.** $\frac{\pi}{2}, \frac{5\pi}{6}$

**d.** $\frac{\pi}{3}, \frac{5\pi}{3}$

Factor: $(2\sin x + 1)(\sin x - 1) = 0$. $\sin x = 1 \Rightarrow x = \pi/2$. $\sin x = -1/2 \Rightarrow x = 7\pi/6, 11\pi/6$.

**30** [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

Solve $\sin(2x) = \cos x$ on $[0, 2\pi)$.

**a.** $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

**b.** $\frac{\pi}{6}, \frac{5\pi}{6}$

**c.** $\frac{\pi}{3}, \frac{2\pi}{3}$

**d.** $\frac{\pi}{2}, \frac{3\pi}{2}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \pi/2, 3\pi/2$.
$\sin x = 1/2 \Rightarrow x = \pi/6, 5\pi/6$.

© ClipSAT — clipsat.org | All Rights Reserved

**31** [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

Solve $\cos(2x) + \cos x = 0$ on $[0, 2\pi)$.

**a.** $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$ ✓

**b.** $\frac{\pi}{6}, \pi, \frac{11\pi}{6}$

**c.** $\frac{\pi}{2}, \frac{3\pi}{2}$

**d.** $\frac{\pi}{3}, \frac{2\pi}{3}$

$2\cos^2 x - 1 + \cos x = 0 \Rightarrow (2\cos x - 1)(\cos x + 1) = 0$. $\cos x = 1/2 \Rightarrow x = \pi/3, 5\pi/3$. $\cos x = -1 \Rightarrow x = \pi$.

**32** [Intermediate] • Target: Double-Angle Formulas

If $\cos\theta = -\frac{7}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{16}{25}$

**b.** $-\frac{4}{5}$

**c.** $\frac{4}{5}$ ✓

**d.** $\frac{9}{10}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(-\frac{7}{25}\right)}{2}} = \frac{4}{5}$.

**33** [Intermediate] • Target: Half-Angle and Power-Reducing Formulas

If $\cos\theta = \frac{23}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $-\frac{1}{5}$

**b.** $\frac{3}{10}$

**c.** $\frac{1}{5}$ ✓

**d.** $\frac{1}{25}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{23}{25}\right)}{2}} = \frac{1}{5}$.

**34** [Intermediate] • Target: Solving Linear Trigonometric Equations

If $\cos\theta = \frac{17}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{1}{2}$

**b.** $-\frac{2}{5}$

**c.** $\frac{2}{5}$ ✓

**d.** $\frac{4}{25}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{17}{25}\right)}{2}} = \frac{2}{5}$.

© ClipSAT — clipsat.org | All Rights Reserved

**35** [Intermediate] • Target: Quadratic-Type and Multiple-Angle Equations

If $\cos\theta = \frac{7}{25}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{9}{25}$

**b.** $\frac{3}{5}$ ✓

**c.** $-\frac{3}{5}$

**d.** $\frac{7}{10}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{7}{25}\right)}{2}} = \frac{3}{5}$.

**36** [Intermediate] • Target: Double-angle formulas

If $\cos\theta = -\frac{1}{8}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $-\frac{3}{4}$

**b.** $\frac{17}{20}$

**c.** $\frac{9}{16}$

**d.** $\frac{3}{4}$ ✓

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(-\frac{1}{8}\right)}{2}} = \frac{3}{4}$.

**37** [Intermediate] • Target: Solving trigonometric equations

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{1}{4}$

**b.** $\frac{1}{2}$ ✓

**c.** $\frac{3}{5}$

**d.** $-\frac{1}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1}{2}\right)}{2}} = \frac{1}{2}$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Half-angle formulas • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{7}{9}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{13}{30}$

**b.** $\frac{1}{3}$ ✓

**c.** $-\frac{1}{3}$

**d.** $\frac{1}{9}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{7}{9}\right)}{2}} = \frac{1}{3}$.

© ClipSAT — clipsat.org | All Rights Reserved

**39** [Challenging] • Target: Equations requiring double-angle substitution • Cross-Lesson: 1.01 Functions and Function Notation

If $\cos\theta = \frac{1}{9}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

**a.** $\frac{2}{3}$ ✓

**b.** $\frac{4}{9}$

**c.** $\frac{23}{30}$

**d.** $-\frac{2}{3}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1}{9}\right)}{2}} = \frac{2}{3}$.

**40** [Challenging] • Target: Applications of double-angle formulas • Cross-Lesson: 1.02 Domain and Range

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point $P$ is given by $3x + 4y = k$. Find the value of $k$.

**a.** **25** ✓

**b.** 30

**c.** 35

**d.** 40

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at $P$. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

**41** [Challenging] • Target: More equation practice • Cross-Lesson: 1.03 Average Rate of Change

On the unit circle, a point $P$ is located at coordinates $(x, y)$. If the angle formed by the positive x-axis and the line segment from the origin to $P$ is $\theta$, what is the value of $x$ in terms of $\theta$?

**a.** $\cos\theta$ ✓

**b.** $\sin\theta$

**c.** $\tan\theta$

**d.** $1$

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle $\theta$ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point $P$ is $\cos\theta$.

© ClipSAT — clipsat.org | All Rights Reserved

**42** [Challenging] • Target: Half-angle formulas for tangent • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Let the complex number $z$ be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose $z$ satisfies the equation $z^3 = -64i$. If $z$ is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of $z$.

**a.** 4

**b.** -4

**c.** $2\sqrt{3}$ ✓

**d.** $-2\sqrt{3}$

We are given the equation $z^3 = -64i$. First, express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). So, $-64i = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Let $z = r(\cos\phi + i\sin\phi)$. Then $z^3 = r^3(\cos(3\phi) + i\sin(3\phi))$. Equating the polar forms, we have $r^3 = 64$ and $3\phi = \frac{3\pi}{2} + 2k\pi$ for some integer $k$. From $r^3 = 64$, we get $r = 4$ (since $r$ is a non-negative real number). Fro.

**43** [Challenging] • Target: Equations with multiple angle types • Cross-Lesson: 2.02 Radian Measure and Arc Length

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 10)

**a.** $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

**b.** $x = \frac{\pi}{2}$ only

**c.** $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

**d.** $x = 0, \pi$

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

**44** [Challenging] • Target: Triple-angle preview • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 4)

**a.** $x = 0, \pi$

**b.** $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

**c.** $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

**d.** $x = \frac{\pi}{2}$ only

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

© ClipSAT — clipsat.org | All Rights Reserved

**45** [Challenging] • Target: Graphical verification of a double-angle identity • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 5)

**a.** $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

**b.** $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

**c.** $x = 0, \pi$

**d.** $x = \frac{\pi}{2}$ only

Using $\cos(2x) = 1 - 2\sin^2 x$: $1 - 2\sin^2 x + \sin x = 0 \Rightarrow 2\sin^2 x - \sin x - 1 = 0$. Factoring: $(2\sin x + 1)(\sin x - 1) = 0$, so $\sin x = 1$ or $\sin x = -\frac{1}{2}$. This gives $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$.

© ClipSAT — clipsat.org | All Rights Reserved